

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5116 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

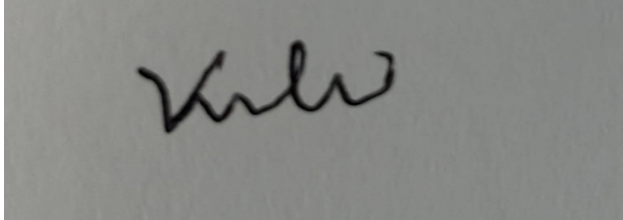
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I _____VARSHA.S/192511358_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in black ink on a light gray background. The signature is stylized and appears to be 'V. K. W.' or similar.

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Design pseudocode for a **stateful firewall system** that tracks active connections and filters packets based on session state. Evaluate how state tracking improves security compared to stateless filtering.
2. Design pseudocode for a **signature-based malware detection system** that scans files and compares them against a known signature database. Evaluate its effectiveness in detecting known threats and its limitations against new attacks.
3. Design pseudocode for a **distributed denial-of-service (DDoS) detection system** using traffic threshold analysis. Evaluate how the algorithm differentiates between legitimate high traffic and attack traffic.

4. Design pseudocode for a **network intrusion detection system using machine learning classification**. Evaluate how feature selection and training data impact detection accuracy.

5. Design pseudocode for a **secure network access control system** that authenticates users and enforces role-based access policies. Evaluate how access control mechanisms enhance overall network security.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to	Mostly clear with minor	Difficult to follow in parts	Poorly structured and unclear

		follow	issues		
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

1. Stateful Firewall System

Pseudocode:

ALGORITHM StatefulFirewall

INPUT: Incoming/Outgoing Packet

DATA: ConnectionTable

FOR each packet DO

 Extract:

 SourceIP, DestinationIP

 SourcePort, DestinationPort

 Protocol

 Connection ← FindConnection(ConnectionTable,

SourceIP, DestinationIP,
SourcePort, DestinationPort, Protocol)

IF Connection exists THEN

IF Connection.State = ESTABLISHED THEN

ALLOW packet

Update Connection timeout

ELSE IF Connection.State = RELATED THEN

ALLOW packet

Update ConnectionTable

ELSE

DROP packet

END IF

ELSE

IF Packet starts a valid new connection THEN

Create new Connection

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    Connection.State ← ESTABLISHED
    Add Connection to ConnectionTable
    ALLOW packet

ELSE
    DROP packet
END IF
END IF

END FOR

END ALGORITHM
```

Evaluation:

A stateful firewall maintains information about active network sessions, whereas a stateless firewall evaluates each packet independently. State tracking improves security because the firewall can determine whether a packet belongs to a legitimate existing connection. Unexpected packets that do not match an active session can be rejected. It also reduces the possibility of unauthorized packets entering an established network connection. However, maintaining connection tables requires additional memory and processing. Overall, stateful filtering provides stronger security and better context-aware packet filtering than stateless filtering.

2. Signature-Based Malware Detection System**Pseudocode:**

ALGORITHM SignatureBasedMalwareDetection

INPUT: File

DATA: SignatureDatabase

Calculate FileHash $\leftarrow$ Hash(File)

IF FileHash exists in SignatureDatabase THEN

 Display "Known Malware Detected"

 Quarantine File

 Generate Security Alert

ELSE

 Scan File contents for known malware patterns

 IF KnownPattern found THEN

 Display "Malware Detected"

 Quarantine File

 Generate Security Alert

 ELSE

 Display "No Known Malware Detected"

END IF

END IF

END ALGORITHM

Evaluation:

Signature-based detection is highly effective against known malware because previously identified malware patterns or hashes can be matched quickly and accurately. It generally produces reliable results when the signature database is regularly updated. Its major limitation is that it cannot reliably detect

completely new malware, zero-day attacks, or heavily modified malware for which no signature exists. Therefore, signature detection should be combined with behavioral analysis, heuristic detection, or machine-learning techniques for improved protection.

3. DDoS Detection Using Traffic Threshold Analysis

Pseudocode:

ALGORITHM DDoSDetection

INPUT: NetworkTraffic

SET NormalThreshold

SET WarningThreshold

SET AttackThreshold

FOR each time interval DO

Collect:

PacketCount

ConnectionCount

SourceIPCount

BandwidthUsage

Calculate TrafficRate

IF TrafficRate $\leq$ NormalThreshold THEN

Status $\leftarrow$ "Normal"

Allow Traffic

ELSE IF TrafficRate $\leq$ WarningThreshold THEN

Status $\leftarrow$ "High Traffic"

Monitor Traffic Closely

ELSE

Analyze:

SourceIP distribution

Connection patterns

Request frequency

Protocol distribution

IF TrafficRate > AttackThreshold AND

TrafficPattern is Abnormal THEN

Status $\leftarrow$ "Possible DDoS Attack"

Rate-limit suspicious traffic

Block highly suspicious sources

Generate Alert

ELSE

Status $\leftarrow$ "Legitimate High Traffic"

Continue Monitoring

END IF

END IF

END FOR

END ALGORITHM

Evaluation:

The algorithm first measures traffic volume and compares it with predefined thresholds. However, high traffic alone does not necessarily indicate a DDoS attack because legitimate events such as a popular website release or online event can also generate large traffic volumes. Therefore, the algorithm additionally examines source distribution, request frequency, connection behavior, and protocol patterns. A DDoS attack generally produces unusual and sustained traffic patterns, while legitimate traffic tends to show more natural user behavior. Adaptive thresholds based on historical traffic can further reduce false positives.

4. Machine-Learning-Based Network Intrusion Detection System

Pseudocode:

ALGORITHM MLBasedIntrusionDetection

INPUT: NetworkDataset

Step 1: Collect network traffic data

Step 2: Extract features:

PacketSize

Protocol

SourcePort

DestinationPort

ConnectionDuration

PacketRate

ByteRate

Step 3: Preprocess data

Remove invalid records

Handle missing values

Normalize numerical features

Step 4: Select important features

Calculate feature importance

Remove irrelevant or redundant features

Step 5: Split dataset into:

TrainingData

ValidationData

TestingData

Step 6: Train MachineLearningModel

Model $\leftarrow$ Train(TrainingData)

Step 7: Validate Model

Tune model parameters

Evaluate using ValidationData

Step 8: Test Model

Predictions $\leftarrow$ Model(TestData)

Step 9: Calculate:

Accuracy

Precision

Recall

F1Score

FOR each new network connection DO

Extract required features

Preprocess features

Result $\leftarrow$ Model.predict(features)

IF Result = "Intrusion" THEN

Generate Alert

Record Intrusion

ELSE

Allow/Monitor Traffic

END IF

END FOR

END ALGORITHM

Evaluation:

Feature selection has a major effect on intrusion-detection accuracy. Relevant features help the model distinguish normal traffic from malicious traffic, while irrelevant or redundant features can increase computational cost and reduce performance. Training data is equally important. A dataset containing diverse and representative normal and attack traffic can improve generalization. If the training data is biased, outdated, too small, or imbalanced, the model may produce false positives or fail to detect certain attacks. Therefore, careful feature selection, balanced training data, regular retraining, and independent testing are important for a reliable IDS.

5. Secure Network Access Control Using RBAC

Pseudocode:

ALGORITHM SecureNetworkAccessControl

INPUT: UserCredentials, RequestedResource

DATA: UserDatabase, RoleDatabase, AccessPolicy

Step 1: Receive username and password

Step 2: Verify credentials

IF credentials are invalid THEN

 Deny Access

 Generate Authentication Failure Alert

 STOP

END IF

Step 3: Identify UserRole

$\text{UserRole} \leftarrow \text{GetRole}(\text{UserDatabase}, \text{Username})$

IF UserRole does not exist THEN

 Deny Access

 Generate Security Alert

 STOP

END IF

Step 4: Determine permissions

Permissions $\leftarrow$ GetPermissions(RoleDatabase, UserRole)

Step 5: Check requested resource

IF RequestedResource is included in Permissions THEN

 Grant Access

 Record Successful Access

ELSE

 Deny Access

 Record Unauthorized Access Attempt

END IF

Step 6: Monitor user activity

IF suspicious activity is detected THEN

 Restrict or terminate session

 Generate Security Alert

END IF

END ALGORITHM

Evaluation:

Role-Based Access Control (RBAC) improves network security by assigning permissions according to user roles rather than giving every user unrestricted access. For example, an administrator can have management privileges while an ordinary user receives only the permissions necessary for normal activities. This follows the principle of least privilege and reduces the damage that can occur if an account is compromised. Authentication verifies the user's identity, while authorization determines what that user is allowed to access. Combining strong authentication, RBAC, logging, and monitoring provides a more secure network access-control system.